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RESEARCH ARTICLE

Large-Scale Simulation of Realistic Cardiac Ultrasound Data With Clinical Appearance: Methodology and Open-Access Database

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ABSTRACT Cardiac ultrasound imaging is widely used in the clinical setting. Deep learning algorithms have shown increased potential in automating routine clinical tasks for improved diagnosis and prognosis. However, the lack of large and diverse training databases hinders their employment on a full scale in clinics. Difficulties associated with creating large training databases include scarcity of clinical data, privacy related data access concerns and labor-intensive ground-truth annotations. To overcome these issues, realistic cardiac ultrasound simulation is considered advantageous. Nevertheless, our state-of-the-art cardiac ultrasound simulation pipeline is limited to small-scale databases particularly because of the underlying computationally complex finite-element heart model. In this paper, we developed a realistic simulation pipeline to facilitate large-scale synthetic cardiac ultrasound datasets. This pipeline utilizes a well-validated and computationally efficient lump-parameter heart model for generating cardiovascular mechanics, and the convolution based ultrasound simulator COLE for short simulation times. We implemented the proposed pipeline to create a large open-access database of 2-D realistic cardiac ultrasound recordings with diversity in the geometries, ultrasound speckle texture and motions of the left ventricle. Quantitative assessment of the simulated data demonstrated that the synthetic cardiac ultrasound frames contained realistic speckle patterns. Moreover, the left ventricular motion in the simulated frames was tracked using the medical imaging tracking toolbox and was shown to closely match the kinematically imposed ground-truth left ventricular motion. The presented pipeline is valuable for creating large and diverse ground-truth training databases. The resulting database is a potential augmentation tool for machine learning-based ultrasound data processing algorithms and made publicly available.

INDEX TERMS Cardiac ultrasound simulation, echocardiography, large-scale simulations, open access, realistic, speckle tracking, strain patterns, synthetic database, ultrasound simulation, validation.

I. INTRODUCTION

Ultrasound (US) imaging is widely used in cardiology owing to its high temporal resolution, cost effective instrumentation, portability, and safety. Echocardiography plays a vital role in detecting and assessing cardiovascular diseases (CVDs). Although acquiring cardiac US images is relatively fast and efficient, routine clinics use manual or semi-automatic tools to measure and assess cardiac morphology, such as cavity volume, and function, e.g. myocardial deformation. This results in an increased time burden and introduces inter and intra-observer variability [1], [2] of morphological

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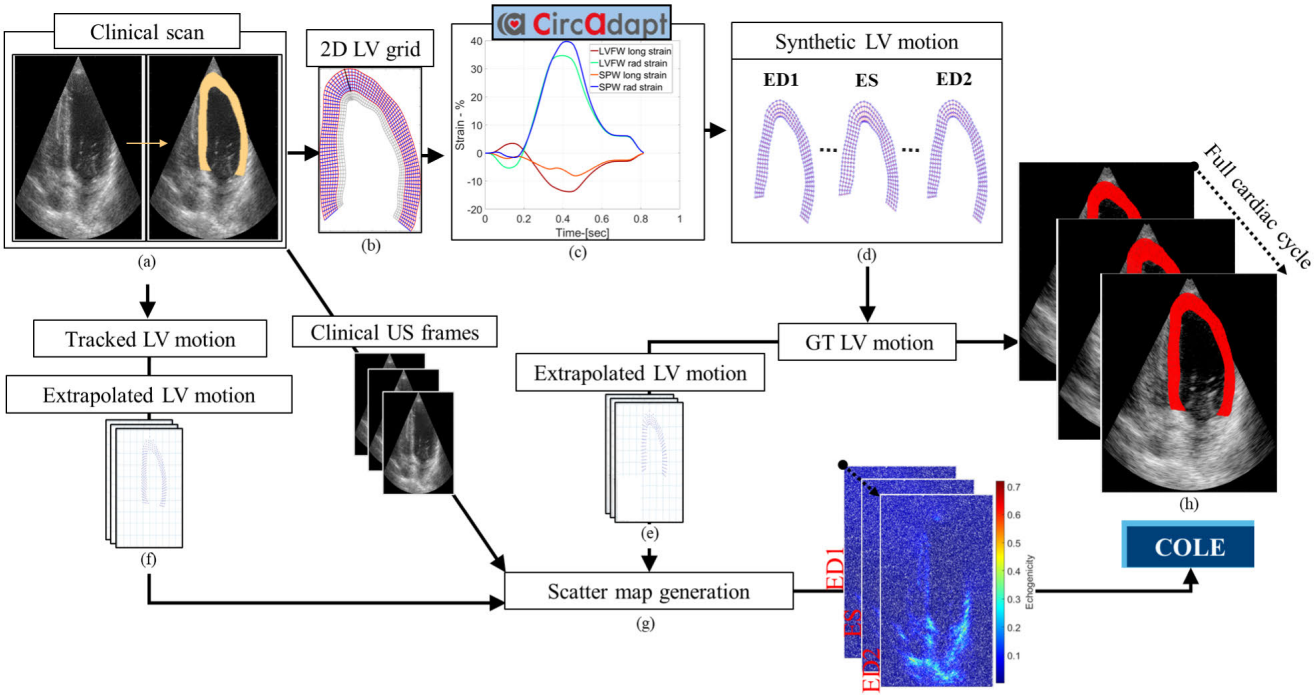

FIGURE 1. Schematic of the proposed realistic cardiac US simulation pipeline. (a) Clinical cardiac US scan with LV segmentation provided in the CAMUS dataset (section II-A). (b) 2D LV grid created from the LV segmentation (section II-C). (c) Radial and longitudinal strain patterns simulated using the CircAdapt model (section II-B). (d) Kinematically imposed synthetic LV motion (section II-D). (e) Extrapolated kinematically imposed LV motion (section II-E). (f) Extrapolated tracked LV motion (section II-E). (g) Scatter map generation scheme (section II-F). (h) Simulated cardiac US frames for the full cardiac cycle with realistic US texture obtained from the clinical scan (section II-G). For visual reference, the synthetic LV motion is superimposed on the simulated frames (in red). The output of the pipeline includes the simulated radio frequency (RF) data, the corresponding scan converted cardiac US frames, the underlying scatter maps, and the GT LV motion as reference providing necessary information to train new machine and deep learning tools.

and deformational indices. Automation in clinical workflow is therefore essential to improve not only the productivity but also the quality and consistency of the diagnosis and prognosis of CVDs.

Deep learning-based algorithms are on the rise in automated general image analysis and computer vision. They have been successfully applied to medical imaging modalities such as magnetic resonance imaging [3] and X-ray imaging [4]. In echocardiography, Convolutional Neural Network (CNN)-based learning algorithms have demonstrated efficacy in performing routine tasks related to classification, segmentation and detection [5], [6], [7], [8], [9]. In Leclerc et al. [10], the largest open-access CAMUS database consisting of 500 clinical US scans was introduced for further development and evaluation of deep learning methods in cardiac US image processing and assessment. Regardless of the aforementioned, potential learning algorithms are yet to reach their full potential in clinics due to their requirement of large labelled training databases to achieve human-level performance.

Major hurdles in curating such medical image databases include the scarcity of clinical data, and tedious and time-consuming ground truth annotations. Additionally, privacy laws restrict accessibility [11]. In this regards, realistic synthetic data generation is the key to enable the construction of large medical image training datasets, overcoming data scarcity. In [12], synthetically generated realistic data is demonstrated to be viable for training machine learning algorithms, substantially increasing the performance of X-ray-based image analysis due to larger training sample size. Synthetic data, as a data augmentation tool, has been used to complement real clinical training datasets, leading to increased performance of various machine learning-based data processing algorithms [13], [14], [15]. These improvements outline the effectiveness of realistic synthetic data in enhancing the performance of machine learning algorithms across different modalities.

In light of the above, ultrasound (US) simulation tools are rapidly gaining importance because of their utility in developing and validating CNN-based learning algorithms for image processing and analysis in the field of echocardiography [16], [17], [18]. Evain et al. [18] demonstrated that increasing the number of simulated cardiac US scans in the training dataset led to a significant improvement in the performance of their motion estimation deep learning algorithm, showcasing the importance of synthetic data as a data augmentation tool with similar findings reported independently by our team [19]. In general, cardiac US image simulation is broadly used ranging from the validation and standardization of speckle tracking based motion estimation algorithms [20], [21] to education purposes [23]. In [20], Alessandrini et al. gave a comprehensive review of the

state of the art in cardiac US simulation. They presented a novel framework for generating realistic synthetic cardiac US recordings. The framework exploited a clinical cardiac US recording to extract the geometry of the heart and realistic speckle texture, and used a finite element (FE) Bestel-Clement-Sorine (BCS) electromechanical (E/M) model [24] to impose a synthetic cardiac motion field. A fast convolution based US simulator (COLE) [25] was used to generate cardiac US data as it offered a good compromise between accuracy and computational complexity. An open access database of 8 3-D [20] and 105 2-D [21] synthetic ultra-realistic cardiac US recordings with corresponding ground truth (GT) cardiac motion was created.

Despite the realism and multifunctional usability of these databases, several shortcomings prevent this framework from being used for creating large databases. Although the underlying cardiac BCS model is state-of-the-art, it is computationally complex and intensive. Additionally, the BCS model requires tedious personalization for simulating pathological hearts, and slight changes in its parameter settings sometimes produce non-physiological wall mechanics limiting the deployment of such frameworks to small-scale databases. Other myocardial kinematic models have been proposed in the literature (e.g. [22]), but they suffer from the same limitation of computational complexity making their use more cumbersome.

In Evain et al. [18], the authors tackled the issue of computational complexity associated with the BSC cardiac FEM model, by extracting a wide variety of left ventricular (LV) motion through manual annotation and resampling the LV endocardial and epicardial borders on clinical scans for the entire cardiac cycle. Although the obtained LV motion fields do not correspond exactly to the actual underlying LV motion fields, they are useful as a relevant data augmentation tool for deep learning-based algorithms. Despite this, the task of manually delineating the myocardial boundaries remains tedious and time-consuming, which limits the resulting dataset to a relatively small scale.

A fast and relatively low-order CircAdapt model has the capability of producing beat-to-beat mechanics and hemodynamics of the cardiovascular system in real time [26], [27], [28]. This multiscale model comprises of various concatenated modules that can be modified based on their relevance and sensitivity to the desired output, making the underlying computations efficient. Unlike FE models of the heart [24], the CircAdapt model uses a simplified spherical geometry to model cardiac walls, namely the right ventricular free wall (RVFW), septal wall (SPW) and the left ventricular free wall (LVFW), enclosing ventricular chambers of the heart, allowing mechanical interactions between the ventricular walls [29]. A unique feature of this model is its inbuilt structural adaptation of cardiac tissue based on the mechanical load, produced by the model itself, to generate physiological stress and strain in the cardiac walls [26]. Despite the simplified geometry, the CircAdapt model has been shown to simulate realistic regional myocardial mechanics and global cardiac pump function for normal as well as various pathophysiological conditions such as ventricular mechanical dyssynchrony [28], [30], [31].

In this paper, we propose a novel realistic cardiac US simulation pipeline that enables the generation of 2D synthetic cardiac US recordings on a large scale. Our overall approach is based on the core concepts outlined in [20] and [21], but it overcomes the limitation related to the dataset size their pipeline can generate. Unlike the manual LV motion estimation strategy used by Evain et al. [18], our method offers an automated and efficient solution. The proposed pipeline employs COLE for fast US simulations, and the CircAdapt model for fast generation of realistic regional myocardial deformation signals labeled with a variety of underlying disease substrates. In addition, clinical scans provided in the CAMUS dataset were used to sample echogenicities for realistic US speckle texture and corresponding annotations were used for diverse left ventricle (LV) geometries. We implemented this pipeline to create an open-access database of 1296 simulated realistic cardiac US recordings along with GT motion of the LV.

This paper proceeds as follows. Section II describes the kinematic US simulation pipeline and its constituents. In section III, we assess the statistical characteristics of the simulated US frames and evaluate the estimated LV motion. Section IV follows with the discussion and section V concludes the paper.

II. PROPOSED SIMULATION PIPELINE

The schematic of the proposed pipeline is shown in Fig. 1 and briefly outlined below. In the initial step we selected a subset of the CAMUS dataset consisting of 50 patients. The selection procedure was mainly based on the scan quality and the clinical indices including ejection fraction (EF) and end diastolic volume (EDV) as explained in section II-A. LV geometries were obtained using the segmentations provided with selected CAMUS scans. The CircAdapt model was used to generate radial and longitudinal global and regional strain patterns for selected CAMUS patients, as explained in section II-B. Two dimensional myocardial grids within the LV boundaries, delineated from the segmentations, were created as described in section II-C. The CircAdapt strain patterns were then used on the respective myocardial grids to kinematically impose 2-D synthetic LV motion for the full cardiac cycle (including first end-diastole (ED), end-systole (ES) and second ED), as explained in section II-D. The synthetic LV motions were then extrapolated to the entire frame. Subsequently, the LV motions were tracked from the respective clinical scans, extrapolated to the entire frame and were temporally aligned to the synthetic LV motion, as explained in section II-E. Using the respective extrapolated motions, echogenicities were sampled from the corresponding time-aligned clinical cardiac US scan to create scatter map for each frame in the entire cardiac cycle, as explained in Section II-F.

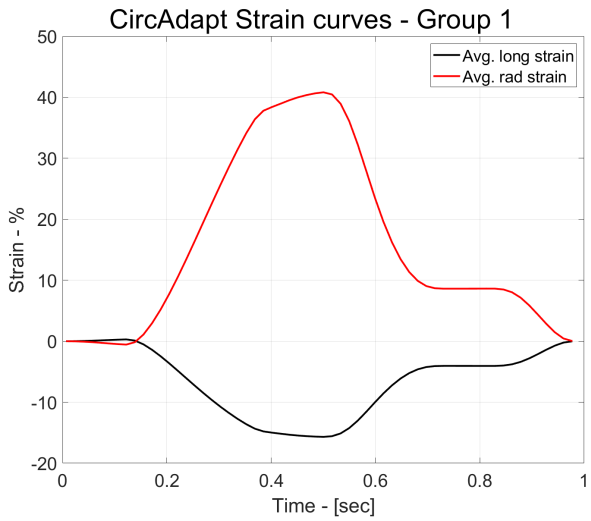

FIGURE 2. Example group 1, global longitudinal and radial strain patterns generated using the CircAdapt model with mitral valve closure as 0 strain reference.

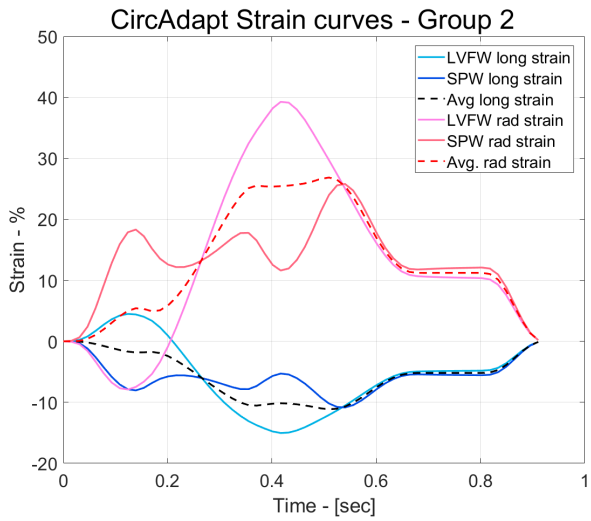

FIGURE 3. Example group 2, regional longitudinal and radial strain patterns, and their average, generated using the CircAdapt model with mitral valve closure as 0 strain reference.

These scatter maps were then utilized in the ultrasound simulator COLE to generate corresponding realistic US frames, as explained in section II-G. The simulated US speckle properties and speckle tracking are detailed in II-H and section II-I, respectively. The pipeline facilitates the simulation of realistic cardiac US sequences with a wide range of fully controlled LV motions in large numbers. Detailed description of the pipeline follows in the subsequent subsections.

A. CAMUS DATABASE

The CAMUS dataset was selected for this work because it is one of the largest, open-access, annotated datasets available, consisting of 500 clinical cardiac US scans covering at least one full cardiac cycle. The US scans were acquired at the University Hospital of St. Etienne, France [10]. Since no selection criteria were adopted during the database curation, typical heterogeneities present in the clinics, in terms of image quality, geometry, and cardiac motion, are consequently preserved. Each scan is provided with annotations at first ED and ES, and clinical indices including EDV and EF.

The clinical scans with reduced EF (EF $<$ 40%) were selected to ensure that the database contained heart failure patients. Moreover, scans with non-physiological clinical indices (EF $<$ 10% and EDV $<$ 60 ml) were discarded. This way, we constructed a subset of the CAMUS database consisting of the clinical scans of 50 patients. Furthermore, LV wall volumes were estimated using the Simpson's bi-plane method using 2D LV geometries at the ED from both two-chamber (2-CH) and four-chamber (4-CH) apical views.

B. CIRCADAPT SIMULATIONS

The CircAdapt model of the human heart and circulation [26], [27], [28] was used to simulate virtual patients with realistic global and regional LV strain patterns. The model is configured as a network composed of a limited number of modules. Each module type represents a system component, such as myocardium, cardiac chambers, valves, blood vessels, and micro-vascular resistances. The chambers representing atria and ventricles are surrounded by walls composed of contractile myocardial tissue. Myocardial tissue is simulated by a nonlinear elastic material. The elastic material harbours myofibers that contract after electromechanical activation. The myofiber stress-strain relation is determined by a three-element Hill model that describes active and passive cardiac myofiber mechanics, incorporating experimental findings on isolated cardiac muscle [26].

The ventricular section of the CircAdapt model is described by the TriSeg model [27]. In this model, three wall segments, representing the LVFW, the SPW (including only the inter-ventricular septum), and the RVFW, meet at a common junction and thereby form both the LV and RV cavity. Interaction between ventricles is included by stating tensional force equilibrium at the junction. The principle of conservation of mechanical work is used to relate global pump mechanics (pressure vs volume relation) of LV and RV to representative myofiber mechanics (stress vs strain relation) in the ventricular walls. The CircAdapt model generates realistic traces of hemodynamic and mechanical variables (such as cardiac and vascular pressures and volumes, flow through cardiac valves, and local myofiber stress vs strain relationship in the constituting walls) as a function of time for the entire cardiac cycle. In this work, however, we have only focused on the myofiber mechanics, in particular the strain patterns of the LV myocardium consisting of the LVFW and SPW.

The Sobol Low discrepancy sequence sampling technique [32] was utilized to generate sets of input parameters that are sensitive to the LV myocardial strain patterns. Each

Limitation? maybe we can improve on this

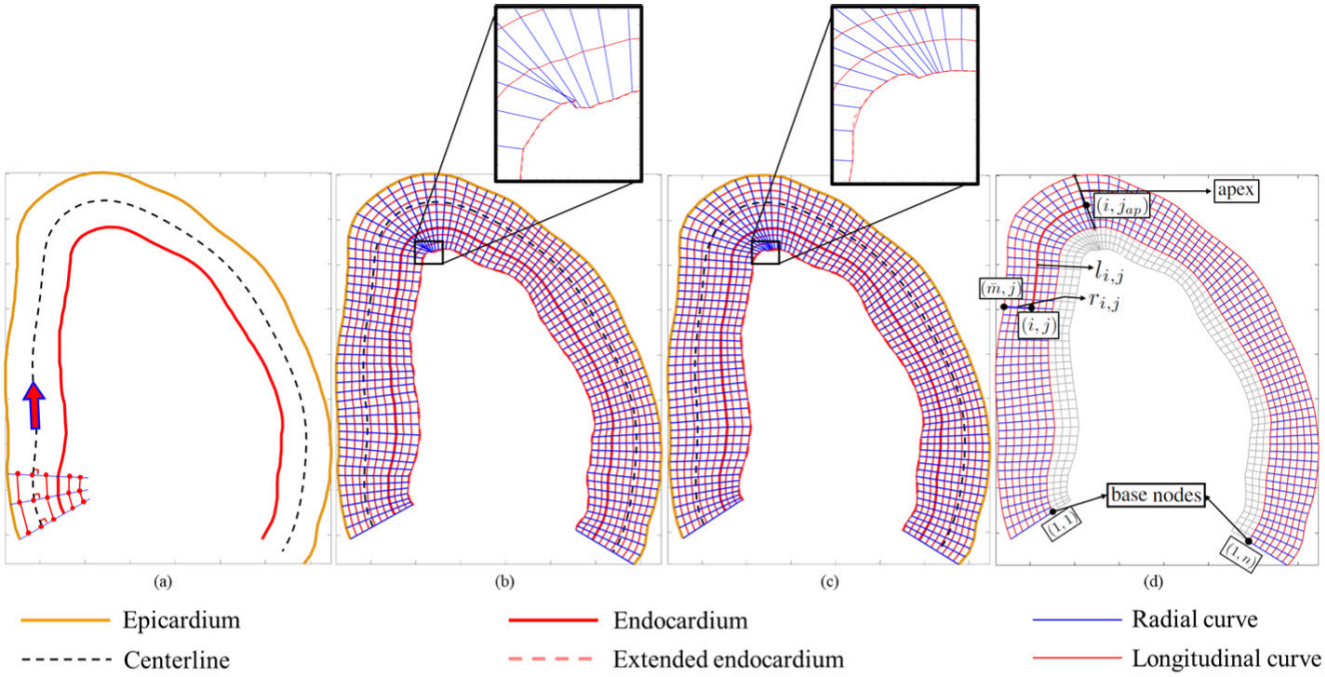

FIGURE 4. Visual representation of grid formation and indexing. (a) Epicardium (orange), endocardium(red) and centerline(dotted black). Extended radial line segments (blue) perpendicular to the centerline are connected along the tangent of the centerline depicted by the red arrow. (b) Depiction of perpendicular radial segments overlap in the high curvature apical region. (c) Myocardial grid after the application of the fix to remove the overlap as explained in section II-C. (d) Indexed myocardial grid. Please note that the myocardial grid was sub-sampled for the readability of the figure.

set contains model parameters describing global cardiac pump function (heart rate, cardiac output, mean arterial pressure and atrioventricular delay), systolic and diastolic tissue properties (contractility, passive stiffness, impaired relaxation). During the sampling process, the values of these parameters are assigned using the predefined ranges for heart failure (HF) patients with reduced EF ($< 40\%$), as shown in the supplementary material, Table 1. Additionally, the generated set also incorporates the LV wall volume which, along with the predefined wall ratio, is used to calculate the input SP wall volumes.

As mentioned previously, each CAMUS patient is characterized by the value of LV wall volume and the EDV. Since the EDV is one of the output parameters rather than an input parameter in the CircAdapt model, a total of 100 sets of input parameters were generated for each patient in the CAMUS subset to ensure the presence of an output EDV that is closest to the patient EDV. Moreover, for that reason, the calculated global LV wall volume of the respective patient was used as an input to the model. The inputs were fed to the CircAdapt model to simulate virtual patients with LV strain patterns of LVFW and SPW and associated clinical indices such as EDV and EF. The virtual patients with EDV values in the $\pm 10\%$ range of the respective CAMUS EDV were selected and the rest were discarded. This ensured the compatibility of the output strain patterns with the geometry of the LV of the respective CAMUS patient.

The filtered simulations were divided into two groups. In the first group, a total of 640 global strain patterns were simulated consisting of average LVFW and SPW strain in the longitudinal and radial direction. An example of the global strain patterns from group 1 is shown in Fig. 2. In the second group, a total of 120 regional strain patterns of the LVFW and SPW in the longitudinal and radial directions were simulated. Notably, in this group, mechanical dyssynchrony was introduced by changing the activation time of the LVFW and SPW. Moreover, these regional strain patterns were also used to estimate average longitudinal and radial strain patterns. An example of the strain patterns from group 2 and their average is shown in Fig. 3.

C. GRID FORMATION

The geometries of the LVs were delineated from the segmentations provided at ED with the scans at apical 2-CH and 4-CH views for each patient in the CAMUS database subset. Firstly, we isolated the endocardium and the epicardium as shown in Fig. 4(a). Secondly, we created a mid curve by connecting mid-points of the shortest line segments between endocardium and epicardium. Subsequently, we re-sampled both endo and epicardium with n numbers of nodes such that the lines connecting them were perpendicular to the mid-curve.

The perpendicular line segments were then extended towards the ventricular cavity to account for the systolic motion of the heart. Each perpendicular segment was then sampled with m number of nodes and connected to adjacent line segments along the tangent of the mid curve to create m longitudinal curves, as shown in Fig. 4(a). However, as shown

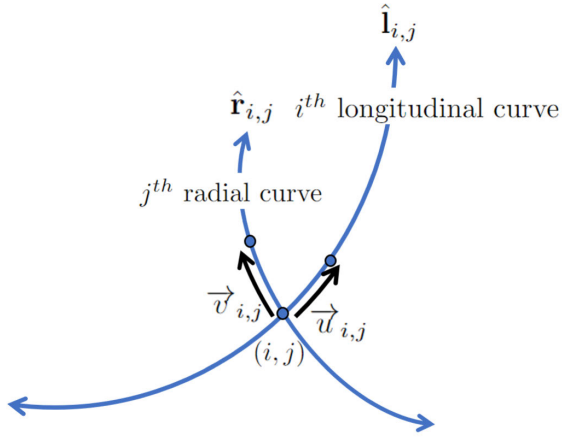

FIGURE 5. Curvilinear coordinate system (l – r) representing the longitudinal unit vector $\hat{l}_{i,j}$ and radial unit vector $\hat{r}_{i,j}$.

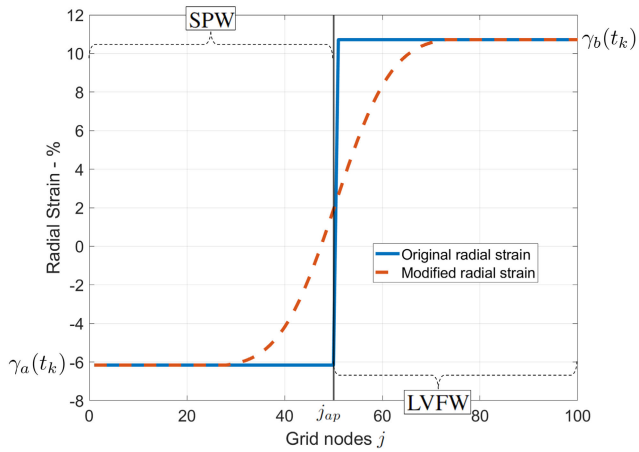

FIGURE 6. Modified radial strain at time t_k: longitudinal grid node index (j) vs radial strain at time t_k, representing the original (blue) and modified (dotted orange) radial strain ($\gamma(t_k)$) distribution. Refer to the text in section II-D for a detailed description.

in Fig. 4(b), this caused perpendicular line segments in the high curvature region, in this case, the apex, to overlap. Later in our simulation, this will result in tissue folding. Therefore, to avoid the grid overlap, we first reduced the number of perpendicular segments to n' such that there was no overlap, which resulted in n' nodes on each longitudinal curve. We then increased the number of nodes back to n, on each longitudinal curve, by interpolating the pitch of the respective curve as a function of its perimeter.

The nodes on each longitudinal curve were connected to its adjacent nodes towards the cavity to create n radial curves. This resulted in a curvilinear grid with m longitudinal and n radial curves, as shown in Fig. 4(c). A point worth noting is that the use of interpolation resulted in bent radial segments in the high curvature region of the LV such as the apex to avoid the grid overlap. Furthermore, to provide clarity, only the myocardial region of the grid is indexed, as shown in Fig. 4(d). The resulting curvilinear myocardial grid consists of $\bar{m} \times n$ number of nodes, where $\bar{m}$ and n were assigned empirically such that the minimum change in myocardial lengths due to the generated strains can be imposed.

In the curvilinear myocardial grid, (i, j) denotes the location of an arbitrary node, where i represents the index of the node on the jth radial curve, and j represents the index of the node on the ith longitudinal curve.

The base was defined by the extreme nodes on the endocardial curve and the apex was defined by a line connecting the epicardial and endocardial nodes which are the furthest away from the basal nodes, as shown in Fig. 4(d). The curvilinear myocardial grid was divided by the apex into two segments: the SPW and the LVFW, as shown in Fig. 4(d). The arc lengths of the longitudinal curves were computed relative to the apex. In particular, $l_{i,j}$ was defined as the arc length of the ith longitudinal curve from the apical node (i, j_{ap}) to an arbitrary node (i, j). Accordingly, the arc lengths of the radial curves were computed relative to the epicardium. Thus $r_{i,j}$ was defined as the arc length of the jth radial curve from the epicardial node $(\bar{m}, j)$ to an arbitrary (i, j) node, also shown in Fig. 4(d).

In this way, the total arc length of the ith longitudinal curve is defined as

$$l_i = l_{i,1} + l_{i,n} \tag{1}$$

where $(i, 1)$ represents the extreme node on the ith longitudinal curve in the SPW region and (i, n) represents the extreme node on the ith longitudinal curve in the LVFW region, as shown in Fig. 4(d). The fraction f_i of the arc length l_i, in the SPW region, is defined as:

$$f_i = \frac{l_{i,1}}{l_i} \tag{2}$$

In a similar manner, the total arc length of the jth radial curve, from the epicardial node $(\bar{m}, j)$ to the endocardial node $(1, j)$, is defined as r_j which is equivalent to $r_{1,j}$.

D. ESTIMATING SYNTHETIC LEFT VENTRICULAR MOTION USING STRAIN PATTERNS

The CircAdapt strain patterns were applied to the respective LV myocardial grids to impose various LV motions/deformation. To enhance the diversity and increase the number of the LV motions, we utilized global strain patterns (Fig. 2), regional strain patterns (Fig. 3), and the average of regional strain patterns (Fig. 3). A detailed description of the methodology to impose LV motion/deformation, involving regional strain patterns, follows. Subsequently, based on this methodology, we describe the methodology using global strain patterns and the averaged regional strain patterns.

A curvilinear coordinate system (l – r) was used to impose the LV motion. In the myocardial grid, each longitudinal curve has its own unit vector $\hat{\mathbf{l}}_{i,j}$, that is always tangent to the ith longitudinal curve for every j^{th} node. Correspondingly, each radial curve has its own unit vector $\hat{\mathbf{r}}_{i,j}$, that is always

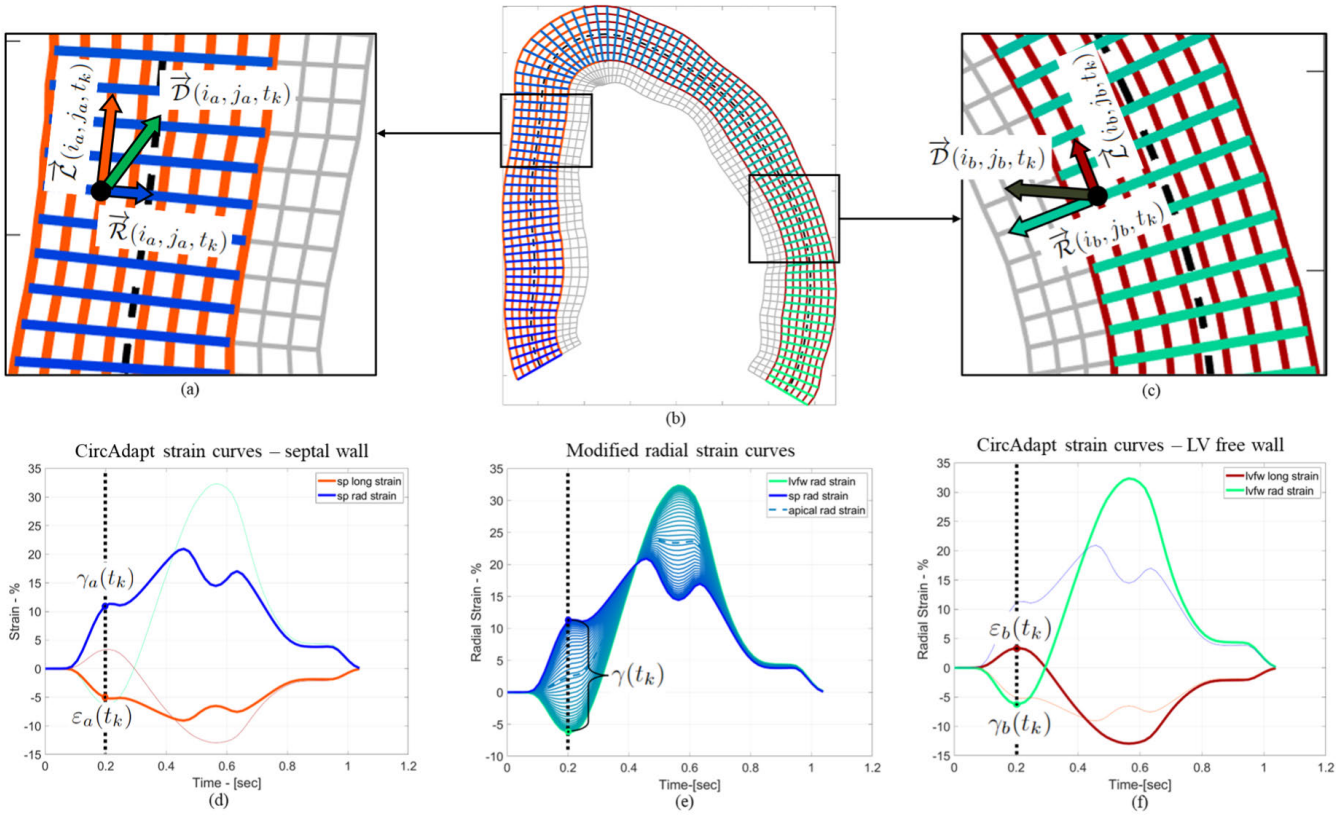

FIGURE 7. **(a-c) Visual representation of the resultant displacement calculations for grid nodes. CircAdapt longitudinal and radial strains simulated for (d) SPW (ε_a, γ_a) and (f) LVFW (ε_b, γ_b). (e) Interpolated radial strains for a gradual transition of radial strain from SPW (γ_a) to LVFW (γ_b). Note that color matching has been used in the figure to illustrate that the strain patterns have been applied to their respective grid curves with the same colors.**

tangent to the jth radial curve for every i^{th} node, as shown in Fig. 5. A point to note is that, unlike the Cartesian coordinate system $(x - y)$, $\hat{l}_{i,j}$ and $\hat{r}_{i,j}$ are not fixed in space but rotate along their respective curve lengths to stay tangential to their respective curves. At each node j on the ith longitudinal curve, displacement in the tangential (or longitudinal) direction is denoted by $\vec{u}_{i,j}$. Similarly, at each node i on j^{th} radial curve, displacement in the tangential (or radial) direction is denoted by $\vec{v}_{i,j}$.

We used SPW longitudinal strain $\varepsilon_a(t_k)$ and LVFW longitudinal strain $\varepsilon_b(t_k)$ to calculate the change in the arc length of the endocardial curve at time t_k where $t_k \in t = [t_1, t_2, \ldots, t_s]$ and t_s is the last time index of the strain patterns. We applied that change in length to each longitudinal curve in the myocardial grid, by displacing the nodes in a linear fashion such that the apical nodes (i, j_{ap}) stay stationary (displacement is zero), and the rest of the nodes move relative to the apical nodes in the longitudinal $(\hat{l})$ direction. In this setting, the longitudinal displacements of all the myocardial nodes at time t_k are given as:

$$\vec{\mathcal{L}}(t_k) = \begin{bmatrix} \vec{\mathcal{A}}(t_k) & \begin{matrix} 0 \\ \vdots \\ 0 \end{matrix} & \vec{\mathcal{B}}(t_k) \end{bmatrix}_{\bar{m}\times n}, \tag{3}$$

$\vec{\mathcal{A}}(t_k)$ represents the longitudinal displacements of all the nodes in the SPW region of the myocardium and is defined as:

$$\vec{\mathcal{A}}(t_k) = \begin{bmatrix} \vec{u}_{1,1}(t_k) & \ldots & \vec{u}_{1,j_{ap}-1}(t_k) \\ \vdots & \ddots & \\ \vec{u}_{\bar{m},1}(t_k) & \ldots & \vec{u}_{\bar{m},j_{ap}-1}(t_k) \end{bmatrix}_{\bar{m}\times(j_{ap}-1)}, \tag{4}$$

$\vec{\mathcal{B}}(t_k)$ represents the longitudinal displacements of all the nodes in the LVFW region of the myocardium and is defined as:

$$\vec{\mathcal{B}}(t_k) = \begin{bmatrix} \vec{u}_{1,j_{ap}+1}(t_k) & \ldots & \vec{u}_{1,n}(t_k) \\ \vdots & \ddots & \\ \vec{u}_{\bar{m},j_{ap}+1}(t_k) & \ldots & \vec{u}_{\bar{m},n}(t_k) \end{bmatrix}_{\bar{m}\times(n-j_{ap})}, \tag{5}$$

where,

$$\vec{u}_{i,j}(t_k) = \begin{cases} \dfrac{\varepsilon_a(t_k)\cdot l_1 \cdot f_i(j_{ap}-j)}{j_{ap}-1} \cdot \hat{l}_{i,j}, & \text{if } 1 \le j \le j_{ap}-1 \\ 0, & \text{if } j = j_{ap} \\ \dfrac{\varepsilon_b(t_k)\cdot l_1 \cdot (1-f_i)(j-j_{ap})}{n-j_{ap}} \cdot \hat{l}_{i,j}, & \text{if } j_{ap}+1 \le j \le n, \end{cases} \tag{6}$$

where f_i is the fraction arc length of the ith longitudinal curve that lies in the SPW region, as defined in (2), and l_1 is the total arc length of the endocardial curve.

Next, to calculate radial displacements of nodes, we first made the transition of the radial strain from SPW ($\gamma_a(t_k)$) region to LVFW ($\gamma_b(t_k)$) region, continuous along the longitudinal direction using a Gaussian-weighted moving average, as shown in Fig. 6. This process generated n radial strain values,

$$\gamma(t_k) = [\gamma_1(t_k), \ldots, \gamma_n(t_k)]_{1\times n}, \tag{7}$$

at time t_k, as illustrated in Fig. 7(e), instead of two discrete values, as shown in Fig. 7(d) and 7(f). Notably, in (7), $\gamma_1(t_k)$ remains equal to $\gamma_a(t_k)$, and $\gamma_n(t_k)$ remains equal to $\gamma_b(t_k)$.

Analogous to the calculation of longitudinal displacements, we used $\gamma_j(t_k) \in \gamma(t_k)$, the modified radial strain for the jth radial curve, to calculate the change in length of the respective radial curve. We applied this change in length to the jth radial curve by displacing its nodes in a linear fashion such that the epicardial node $(\bar{m}, j)$ remained stationary while the remaining nodes moved relative to the epicardial node, towards the cavity. This operation is repeated $\forall j \in [1, \ldots, n]$. In this setting, the radial displacements of all the myocardial nodes at time t_k are given as:

$$\overrightarrow{\mathcal{R}}(t_k) = \begin{bmatrix} \overrightarrow{v}_{1,1}(t_k) & \ldots & \overrightarrow{v}_{1,n}(t_k) \\ \vdots & \ddots & \vdots \\ \overrightarrow{v}_{\bar{m},1}(t_k) & \ldots & \overrightarrow{v}_{\bar{m},n}(t_k) \end{bmatrix}_{\bar{m}\times n}, \tag{8}$$

where,

$$\overrightarrow{v}_{i,j}(t_k) = \begin{cases} \dfrac{\gamma_j(t_k)\cdot r_j\cdot(\bar{m}-i)}{\bar{m}-1}\cdot \hat{r}_{i,j}, & \text{if } 1 \le i \le \bar{m}-1 \\ 0, & \text{if } i = \bar{m}, \end{cases} \tag{9}$$

where $\gamma_j \in \gamma$ is radial strain for j^{th} radial curve and r_j is the total arc length of the jth radial curve from the epicardial node to the endocardial node.

As shown in Fig. 7(a-c), to get the resulting displacements, we added longitudinal displacements $\overrightarrow{\mathcal{L}}(t_k)$ and corresponding radial displacements $\overrightarrow{\mathcal{R}}(t_k)$:

$$\overrightarrow{\mathcal{D}}(t_k) = \overrightarrow{\mathcal{L}}(t_k) + \overrightarrow{\mathcal{R}}(t_k), \tag{10}$$

Imposing 2D LV motion using global strain patterns from group 1 (Fig. 2(a)) was performed by setting regional strain values to be equal to the global strain values, i.e., $\varepsilon_a = \varepsilon_b = \varepsilon$ and $\gamma_a = \gamma_b = \gamma$. Similarly, imposing the LV motion using the average of regional strain patterns was done by setting the longitudinal (ε) and radial (γ) strains in the SPW and LVFW equal to the average of the regional strain patterns i.e. $\varepsilon = \frac{\varepsilon_a+\varepsilon_b}{2}$, and $\gamma = \frac{\gamma_a+\gamma_b}{2}$.

E. EXTRAPOLATED LEFT VENTRICULAR MOTION AND TEMPORAL ALIGNMENT

Sampling echogenicities from the clinical scan requires temporal alignment between the generated strain patterns and the clinical scan, as well as the displacement fields across the entire image domain.

Resultant displacements as defined in (10), describe motion of myocardial grid nodes (q) as $q(t) = q + \overrightarrow{\mathcal{D}}(t)$. To compute the motion outside the LV, $\overrightarrow{\mathcal{D}}$ was extrapolated using thin plane splines (TPS) [33] to the entire image domain, denoted as $\overrightarrow{\mathfrak{D}}$, with $q(t)$ used as control points. In a standard 4 or 2-CH view cardiac US scan, speckle motion between the LV apex and probe is limited by the presence of the skin layer, which exhibits minimal movement, or due to artifacts such as near-field clutter. Therefore, to restrict the motion between the LV apex and the probe in the extrapolated displacement field $\mathfrak{D}$, additional stationary control points were added between the US probe and the LV apex.

The LV motion from the corresponding clinical US scan was tracked using the medical imaging tracking toolbox (MITT) [34]. Similar to the extrapolation of the synthetic LV motion, the tracked LV motion was extrapolated to the entire frame as well.

Furthermore, since the cardiac cycle of the clinical US scan was different than the CircAdapt strain patterns, the main cardiac events (i.e. first ED, ES and second ED) were used for temporal synchronization. First ED and second ED were matched by downsampling the strain patterns, and the ES event was matched by stretching the time axis of the clinical US scan using linear interpolation. This resulted in non-integer temporally matched frames of the clinical US scan.

In the next section, these extrapolated LV motions and the respective temporally aligned clinical US scan are used to create scatter maps for the entire cardiac cycle.

F. SCATTER MAPS GENERATION

A scatter map consists of a collection of US scattering sites and their corresponding scatterer amplitudes, also referred as echogenicity. These amplitudes represent the intensity of the US backscattered towards the probe. As explained in the following section, the US simulator uses the scatter map to output corresponding US data. To give simulated US speckles a realistic texture, the scatterer amplitudes were sampled from clinical scans. Since the dynamic range of the scans was not available, we processed each clinical scan frame by applying gamma compression to compensate for the compression typically applied in clinical US scanners. Subsequently, the compressed frames were multiplied by a Normally-distributed random matrix $\mathcal{N}$ ranging from 0 to 1 to introduce some random variation in the amplitudes [18]. For any frame f, this process can be expressed as

$$\mathcal{C}_f = (I_f/255)^{1/\Gamma}\cdot\mathcal{N} \tag{11}$$

where $\mathcal{C}$ is the estimated amplitudes, I_f represents the intensities in the B-mode frame f of the clinical US scan, and Γ is the gamma coefficient. Due to the variability in the image quality and lack of information about the degree of

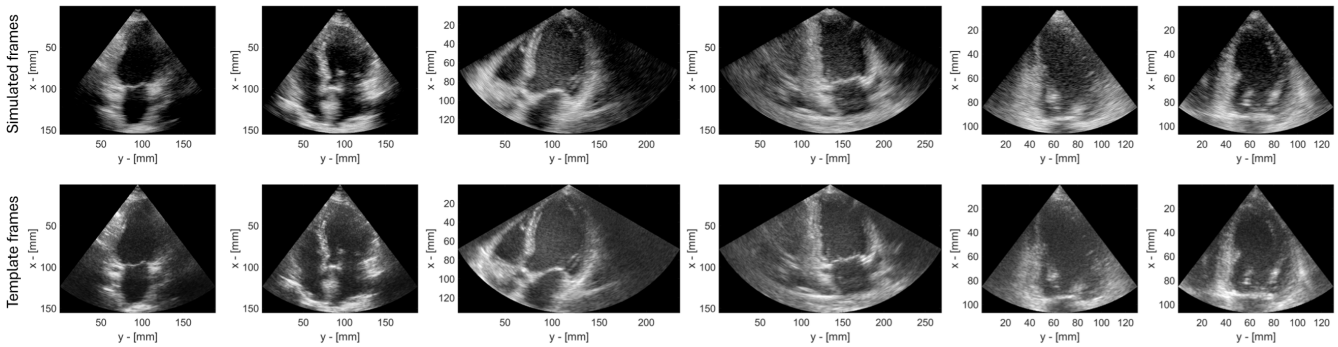

FIGURE 8. Sample ED frames from simulated realistic cardiac US recording (first row) and corresponding template frames used to sample the echogenicity.

compression, gamma coefficients (Γ) in (11) were estimated manually. A scan was randomly selected and empirically assigned a $\Gamma = \Gamma_{ref}$ that visually exhibited good contrast in the simulated US frame. Subsequently, histogram matching with the selected scan was performed to estimate gamma coefficients for the remaining clinical scans in the CAMUS subset. This may result in an overall different contrast in the simulated US frames compared to the clinical scan. However, we consider this method sufficient, as we are not trying to exactly replicate the texture but rather achieve a realistic approximation.

To begin, N random scatterers were uniformly distributed on the ED-frame (i.e., the first frame $f = 1$). The density of these random scatterers was chosen to be 18 scatterers per minimum resolution cell to ensure realistic speckle statistics. The minimum resolution cell was defined using axial (i.e., $\frac{\text{spatial pulse length}}{2} = 0.51$ mm) and minimum lateral resolution (i.e., -6dB minimum lateral beam width $=$ 0.83 mm). These scatterers were then divided into coherent scatterers, which lie on the LV myocardium and incoherent scatterers, which lie in the rest of the background. The coherent scatterers were propagated using the extrapolated displacement field $\mathfrak{D}$, while their amplitudes were sampled from $\mathcal{C}_{f_{ED}}$ and kept constant throughout the cardiac cycle. This was done to maintain speckle coherency within the myocardium. On the other hand, the incoherent scatterers were also propagated using the extrapolated displacement field $\mathfrak{D}$, but their locations were randomly defined in each frame, while their amplitudes were sampled from the time aligned frames of the respective clinical scan. The random shuffling allowed for speckle incoherency in the background throughout the entire cardiac cycle.

Further, to avoid unrealistically high speckle coherence within the myocardium, coherent and incoherent scatterers were merged allowing for some incoherent scatterers to be present in the LV myocardium and vice versa, as described in [21]. This mitigated the edge artifacts that arise due to the sudden change in the speckle coherency between the myocardium and background. This process was repeated for each simulated frame to create scatter maps for the entire cardiac cycle.

TABLE 1. Ultrasound imaging system properties.

Probe parameters	
Number of elements	64
Central frequency	3 MHz
Sampling frequency	50 MHz
Speed of sound	1540 m/s
Element width	0.375 mm
Element height	15 mm
Transmit focus	[50 - 100mm]
Transmit Apodization	Hanning
Probe Width	24 mm
Transmit pulse	
Waveform	Gaussian
Cycle	2
-6dB relative bandwidth	60%

G. ULTRASOUND SIMULATION

The COLE US simulator generates beamformed RF data through multiple 1-D convolutions between the transmit pulse and scatter distribution function [25]. A scatter distribution function comprises the projection of the US scattering sites onto an image line, their respective amplitudes, and projection factors. An image line is defined by its lateral angle with respect to the probe center, which ranges within the pre-defined opening angle. Projection factors are determined using either an analytical expression or a lookup table (LUT) to account for probe sensitivity.

We setup COLE by defining the properties of the imaging system using typical clinical settings for echocardiography, as shown in Table 1. To accommodate various focal depths (ranging from 50 mm to 100 mm) in the simulation of cardiac US recordings, multiple realistic beam profiles for an apodized 64-elements phased array transducer (as defined in Table 1) transmitting a 3 MHz Gaussian pulse with a -6 dB relative bandwidth of 60% were generated using Field II [35]. Each beam profile was simulated on a grid of 2291 by 1000 points within a maximum opening angle 120-degree-wide and depths ranging from 4 mm to 200 mm, with a fixed focus in transmit and dynamic focusing in receive. The fixed transmit focus varied from 50 mm to

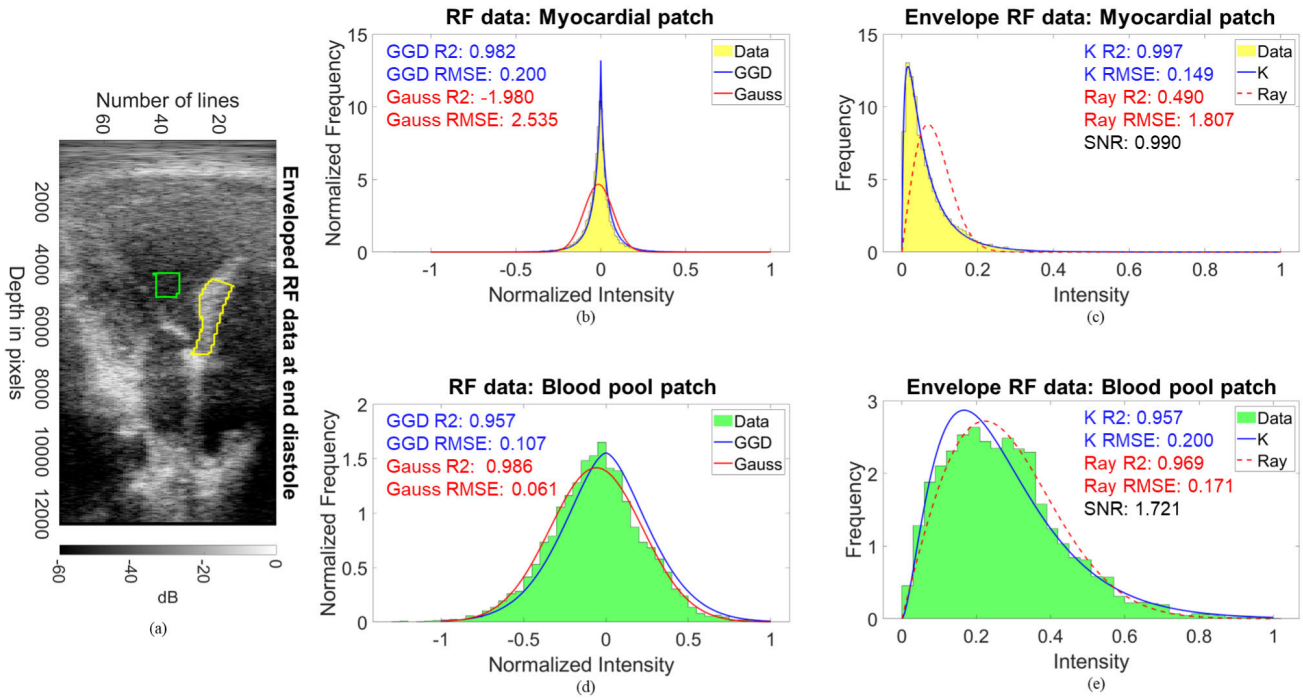

FIGURE 9. Speckle statistics of an example simulated cardiac US frame at ED are shown in this figure. (a) displays envelope RF data at ED that has been log compressed for visualization. The yellow patch represents a region from the LV myocardial tissue, while the green patch represents a region from the blood pool LV cavity. RF data histograms of the myocardial and blood pool patches are shown in (b) and (d), respectively. Envelope RF data histograms of the myocardial and blood pool patches are shown in (c) and (e), respectively. Each histogram plot includes fits with different probability distribution functions, along with the respective RMSE and R-squared values. GGD stands for the generalized Gaussian Distribution, Gauss stands for the Gaussian distribution and Ray stands for the Rayleigh distribution.

TABLE 2. Quantitative analysis of simulated US data.

	SNR	Gaussian		Generalized Gaussian		K-distribution		Rayleigh distribution	
		Beamformed RF data				Envelope data			
		RMSE	R^2	*RMSE*	R^2	*RMSE*	R^2	*RMSE*	R^2
Blood pool	1.62 ± 0.14	0.10 ± 0.05	0.97 ± 0.02	0.12 ± 0.02	0.96 ± 0.01	0.15 ± 0.04	0.97 ± 0.02	0.30 ± 0.13	0.90 ± 0.07
Myocardial tissue	1.17 ± 0.26	1.77 ± 0.76	-1.61 ± 0.98	0.18 ± 0.05	0.97 ± 0.01	0.15 ± 0.08	0.99 ± 0.01	1.14 ± 0.61	0.61 ± 0.23

Average *RMSE* and R^2 values are computed over 156 ED frames. *RMSE* represents root mean square error $\pm$ standard deviation. R^2 represents R-Square value $\pm$ standard deviation. SNR represents the signal-to-noise-ratio defined as the mean over the standard deviation of the intensity in the investigated region.

100 mm in a step of 2.5 mm, resulting in 21 realistic beam profiles.

For each simulation of cardiac US recording, we selected a beam profile with a transmit focal depth closest to half of the total acquired scan depth of the template clinical US scan to be loaded as a LUT. The opening angle was also calculated from the respective template clinical US scan, with an upper limit set to 120 degrees. This process was repeated for all the simulations.

Subsequently, scatter maps, selected LUT, total scan depth, opening angle, and a 3 MHz Gaussian pulse of two cycles with a -6 dB relative bandwidth of 60% were used as input to COLE to simulate beam-formed radio frequency (RF) data for the entire cardiac cycle. The simulated RF data per frame contained one RF line per degree of opening angle. These RF lines were envelope-detected, scan-converted from polar to Cartesian coordinates, and log-compressed to a dynamic scale of 60 dB to create corresponding B-mode frames of 1214 pixel $\times$ 1000 pixel with a pixel size of 0.184 mm $\times$ 0.184 mm, equivalent to $0.36\lambda \times 0.36\lambda$. The number of frames in the resulting cardiac US recordings matched the frames in the corresponding template clinical scans, ranging between 28 and 100 frames per cardiac cycle.

H. EVALUATION OF THE SPECKLE PROPERTIES OF THE SIMULATED ULTRASOUND DATA

The statistical characteristics of the simulated US frames were quantitatively assessed by analyzing the intensity distribution of the simulated US frames. To evaluate these characteristics, we computed histograms of the patches within the myocardial tissue (yellow) and blood pool (green), as shown in Fig. 9(a). The myocardial patch was created in Cartesian coordinates using a part of the respective LV geometry. Similarly, the blood pool patch was also created in Cartesian coordinates by defining a square region of $\sim$10mm $\times$ 10mm. These patches were then converted to polar

coordinates to sample RF amplitudes from the respective regions. The histograms were computed for both simulated RF data and envelope data before downsampling to ensure a high number of samples (>7000 samples).

Due to the variability in the US speckle patterns in the simulated cardiac US data, we selected 17 virtual patients from the CAMUS subset and performed the same histogram analysis for the respective 156 simulated realistic cardiac US ED frames.

I. SPECKLE TRACKING OF THE SIMULATED ULTRASOUND DATA

To demonstrate the validity and usability of the generated cardiac US data, we employed the MITT toolbox to track the LV from the simulated US recording and hence obtain the LV motion. This tracked LV motion was then compared with the corresponding kinematically imposed GT LV motion.

Firstly, a tracking structured grid object was created using the LV segmentations provided with the scans in the CAMUS subset. The grid object consisted of 50 radial line segments equally spaced on the endocardium and epicardium. Each radial line segment was divided into 5 equally spaced nodes resulting in a grid of 250 nodes, as shown in Fig. 10(a). This structured grid object was fed to MITT along with the corresponding simulated US frames. Each node q of the grid object was tracked, and its absolute distance d_q from the corresponding ground truth node, as shown in Fig. 10(b-c), was calculated at each time step $t_k \in t$. Further, for each node q, d_q was averaged over the entire duration of the cardiac cycle and was defined to be the error:

$$\bar{d_q} = \frac{\sum_{k=1}^{p} d_q(t_k)}{p}, \tag{12}$$

where p is the total number of time steps, $k = 1$ corresponds to the first ED and $k = p$ corresponds to the second ED which also denotes the end of a cardiac cycle.

III. RESULTS

A. OPEN-ACCESS DATASET

Examples of simulated realistic cardiac US frames at ED for various shapes of the LV are shown in Fig. 8. We created a database of 1296 realistic cardiac US recordings including diverse shapes of the LV, and a wide range of LV motion/deformation, and realistic US speckle texture directly sampled from the clinical cardiac US scans. Among these, 1092 US recordings were generated using group 1 global strain patterns and group 2 averaged regional strain patterns, while the remaining 204 US recordings were created using group 2 regional strain patterns. For comprehensive visualization, we have provided example videos of the simulated realistic cardiac US recordings, generated using regional and global strain patterns, in the supplementary material.

The created dataset encompasses simulated realistic cardiac US frames (both in 2-CH and 4-CH views), along with corresponding RF data, scatter maps, GT displacement maps, and underlying CircAdapt strain patterns. The entire dataset is made publicly available and can be accessed using the following url [36]: https://doi.org/10.48804/Q1VEX8.

B. HISTOGRAM ANALYSIS OF SIMULATED ULTRASOUND DATA

Fig. 9 illustrates the speckle statistics of an example simulated cardiac US frame at ED, including the histograms of RF data from myocardial and blood pool patches, alongside their corresponding fits to various statistical distributions. The histogram of the RF data from the myocardial patch, as shown in Fig. 9(b), exhibits a good fit to the generalized Gaussian distribution based on the root mean square error (RMSE) and R-Squared values. Similarly, the histogram of the envelope data from the myocardial patch, as shown in Fig. 9(c), exhibits a good fit to the K-distribution.

Correspondingly, the histogram of the RF data from the blood pool patch, as shown in Fig. 9(d), demonstrates a good fit to both the Gaussian and generalized Gaussian distributions, while the histogram of the envelope data from the blood pool patch shows a good fit to the Rayleigh distribution.

The overall mean RMSE, R-squared, and SNR values corresponding to the selected 156 recordings from the simulated dataset are summarized in Table 2. Both the Gaussian and generalized Gaussian distributions seem to give a good fit to the RF histogram from the blood pool patch. The generalized Gaussian distribution is a good fit to the RF histogram from the myocardial patch. The K-distribution is a good fit to the enveloped RF histograms from both the myocardial and blood pool patches. Overall, the mean SNR of the blood pool and myocardial patches was 1.62 ± 0.014 and 1.17 ± 0.26, respectively.

C. EVALUATION OF THE GROUND TRUTH MOTION

The tracking error, as defined in (12), was calculated for all nodes across multiple US recordings and was found to have a mean of 0.43 ± 0.50 mm. For a comprehensive visualization of the tracking performance, we have provided example videos in the supplementary material.

IV. DISCUSSION

Our new realistic cardiac US simulation pipeline leverages the numerical stability, short simulation time, and realism of the CircAdapt heart model, along with the computational efficiency and fast RF data generation of the convolution-based US simulator COLE, to enable large-scale simulation of realistic cardiac US recordings. The proposed pipeline was used to generate a large database of 1296 2-D cardiac US recordings, including 2-CH and 4-CH views. Particularly, since the CircAdapt model deals with suitable parameters that are pertinent to simulating desired abnormalities, extensive simulation of strain patterns for a heart failure population was made possible. Moreover, the simulated cardiac US recordings are diverse in terms of the LV geometries and US speckle texture as a result of clinical heterogeneities present

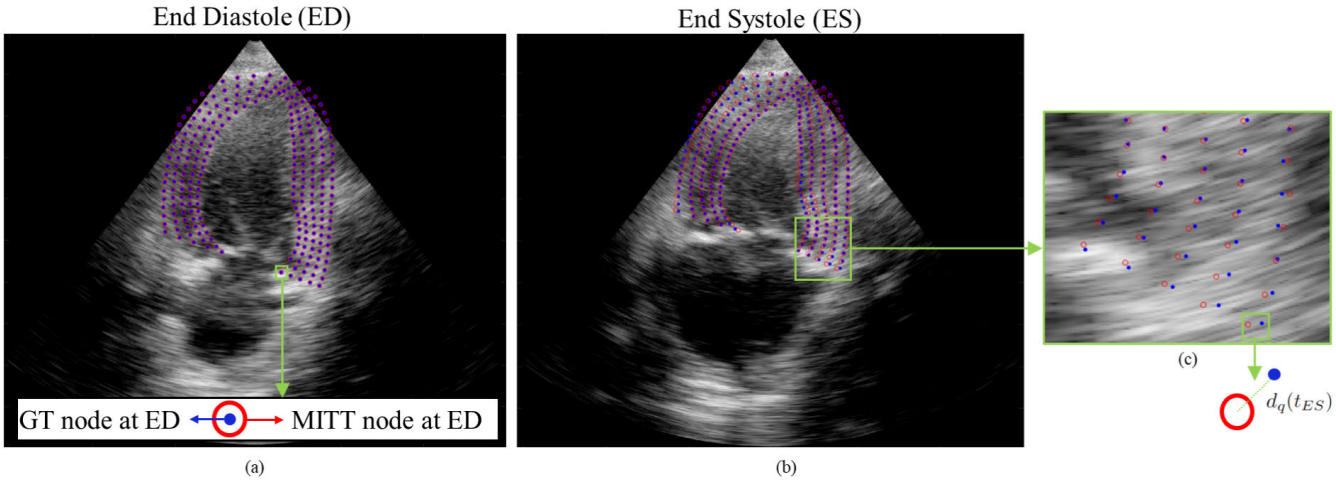

FIGURE 10. Structured grid nodes selected from the myocardial grid for speckle tracking. (a) The red circles are tracked using MITT and the respective blue nodes are displaced by the kinematically imposed GT cardiac motion/deformation. (b-c) Tracked nodes (red) and GT nodes (blue) at the ES. The average absolute distance between the red circles and corresponding blue nodes for the entire duration of the cardiac cycle is defined as the error in motion estimation. Refer to the text in section III-C for a detailed description.

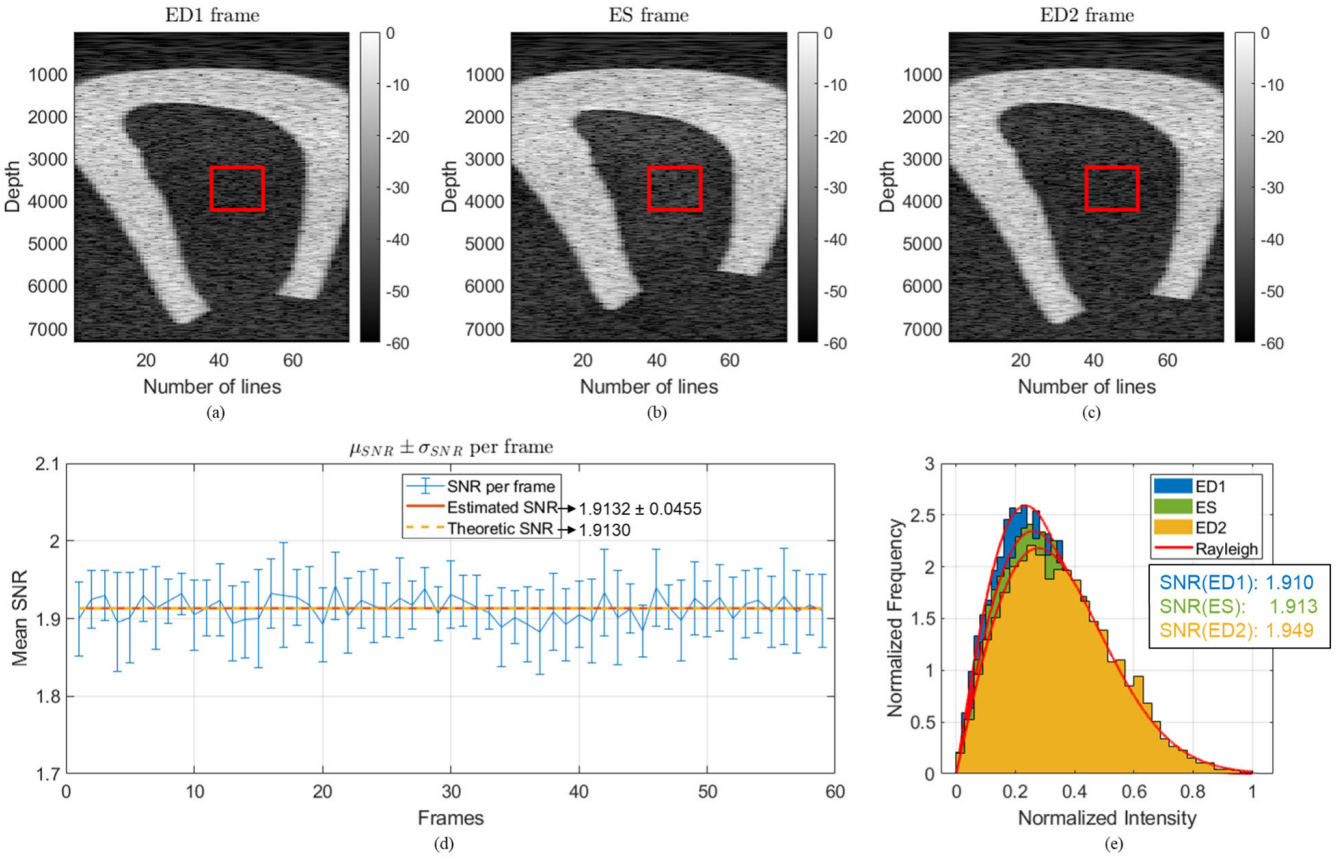

FIGURE 11. US speckle analysis of simulated cardiac US data with echogenicities sampled from Gaussian distributions. (a), (b) and (c) Envelope and log-compressed RF data at first ED, ES and second ED, respectively. The red square patch represents a region from the LV blood pool cavity. (d) Evolution of mean SNR over time (frames). (e) Histogram of the envelope data from first ED, ES and second ED evaluated on the blood pool patch, together with their fit to a Rayleigh distribution.

in the CAMUS database, and cardiac motion due to the use of Sobol-Low Discrepancy sequence in simulating CircAdapt strain patterns for heart failure with reduced EF population.

In contrast, the previous state-of-the-art cardiac US simulation pipeline [21] created an open access database of 105 2D cardiac US sequences, limited in size due

to the use of the computationally complex E/M cardiac model. In [18], Evain et al. avoided this issue by manually annotating the LV contours from echocardiographic scans to generate synthetic motion. While this process results in many global LV deformations, it requires manual delineation of the LV on each frame for the entire cardiac cycle. Consequently, the resulting synthetic cardiac US database is limited in size due to the tedious and time-consuming nature of this task. Additionally, errors in contouring can lead to un-physiologic synthetic motion. Our proposed pipeline successfully overcomes these limitations by generating physiologic GT LV deformation using the CircAdapt model, enabling the creation of significantly larger and more diverse cardiac US databases.

The statistical distribution of the US image histogram is primarily influenced by the number of scattering sites per resolution cell of the US system [37]. Densely sampling echogenicity from a clinical scan allows us to inherit the underlying statistical characteristics of the US speckle. To ensure the clinical relevance of the realistic US speckle texture in the simulated cardiac US recordings, we used a high scatterer density to sample the echogenicity from the respective clinical scans.

Notably, the beamforming steps, including dynamic focusing and apodization, may result in the minimum −6dB beamwidth at depths different from the assigned focal depths. Additionally, the background scatterers are temporally incoherent but their density changes according to the extrapolated LV motion. Therefore, we chose a high scatterer density of 18 scatterers per minimum resolution cell to ensure that there are at least 11 scatterers per resolution cell in the entire region across the frames [38].

The histogram analysis on an example simulated cardiac US frame at ED, as shown in Fig. 9, indicates that the US speckles within the myocardial patch contain structural components which make the scattering deviate from pure Rayleigh scattering, hence the better fit to the generalized Gaussian and K-distributions. These distributions indicate a structural component to the scattering process, which is commonly observed in the tissue region in clinical US recordings [39], [40], [41]. Similarly, the RF and envelope data histograms from the blood pool patch gave good fits to the Gaussian and Rayleigh distribution, respectively. Although these distributions may suggest fully developed speckle [37], [38], [42], [43], the signal-to-noise ratio (SNR) of the envelope data was calculated to be 1.721, which is less than the expected SNR of a Rayleigh-distributed data (i.e., 1.913, as reported in [37]).

After conducting the histogram analysis for 156 simulated ED US frames, as summarized in Table 2, it is evident that the US data from the myocardial patch exhibit speckle characteristics influenced by the myocardial tissue structure. However, the US speckles from the blood pool patch also exhibit statistical characteristics of speckles influenced by underlying tissue structures, as on average the K-distribution gives a good fit for the envelope data histogram, with a mean SNR of 1.62 ± 0.014. In clinical recordings, the blood pool US speckles are typically fully developed and characterized by the Rayleigh distribution, leading to an SNR of 1.913 [37], [38], [42], [43]. We hypothesize that the presence of underlying tissue structures, such as the papillary muscles, and typical echocardiographic artifacts in the blood pool, leads to the deviation of the speckle statistics from the Rayleigh to sub-Rayleigh regime (SNR < 1.913). Given the noisy nature of the clinical scans in the CAMUS dataset used to sample the echogenicities, isolating the blood pool patch from these surrounding structures and artifacts is challenging.

To confirm our hypothesis, we performed an additional analysis where we simulated cardiac US data by modeling the echogenicities from normal random distributions. We sampled the echogenicities of the foreground (myocardial) and background scatterers from normal random distributions with means of 0.8 ± 0.1 and 0.0125 ± 0.0025, respectively. These mean echogenicity values were chosen to achieve $\sim$ 35dB contrast between the myocardium and the blood pool region as typically observed in the clinical recordings. The myocardial and background scatterers were then propagated using the extrapolated LV motions, where the myocardial amplitudes remained constant and background amplitudes were randomly shuffled in each frame. This resulted in the RF data for a full cardiac cycle (Fig. 11 (a-c)). The speckle in the blood pool region, denoted by the red square patch centered at the focus in Fig. 11 (a-c), was quantitatively analyzed for the entire cardiac cycle. For each frame, the RF data were simulated 10 times and the mean SNR of the blood pool region was estimated. The evolution of the mean SNR with frame index is shown in Fig. 11 (d), and the histograms of the blood pool region from three time events (ED1, ES, ED2) are shown in Fig. 11 (e). Overall, the mean SNR was estimated to be 1.9132 ± 0.0455 which is close to the Rayleigh SNR of 1.9130. These results confirm that simulated US speckles in the blood pool region are fully developed, and therefore suggest that the deviation in SNR in the simulated realistic cardiac US data is due to the presence of papillary muscles and artifacts in the blood pool cavity of the template clinical scans.

To ensure that the simulated LV motion corresponded to the kinematically imposed motion, the LV myocardial motion was tracked from the simulated realistic cardiac US data using MITT. The tracked motion was then compared with the kinematically imposed GT LV motion and a sub-millimeter displacement error (0.43 ± 0.50 mm) was observed between the two, as shown in section III-C. The relatively large standard deviation of the tracking error with respect to its mean is due to the fact that, for many simulated recordings, the tracked LV is outside of the sector. Furthermore, the speckle tracking is made challenging by the mixing of background incoherent scatterers with myocardial coherent scatterers, as explained in section II-F. Despite this, the tracking evaluation demonstrated that the simulated speckles could be tracked with a reasonable accuracy. Therefore, the dataset can be used for the analysis and validation

of various tracking techniques, including speckle tracking-based algorithms. The entire set of simulated dataset is made publicly available.

A point to note is that the visual motion in the simulated cardiac US recordings of the background structures, such as the atria and valves, arises from the temporal intensity variation and does not correspond to the extrapolated LV motion. Although, the ground truth motion of these surrounding structures is not available, their rendered motion is visually realistic.

Our simulation framework suffered from several limitations and theoretical approximations, which may have an effect on the realism of the generated cardiac US data. Firstly, although the CircAdapt model produces beat-to-beat realistic mechanics and hemodynamics, it utilizes a simplified spherical geometry to model the ventricles. As a consequence, it does not provide a spatial distribution of strain patterns. Therefore, to apply the generated strains on the LV myocardial grids, a linear spatial distribution of the corresponding magnitudes of myocardial length changes was implemented within the respective segments in both radial and longitudinal directions. Thus, the strain imposed on the LV grid represents an effective GT LV deformation that can deviate from the strain patterns generated by CircAdapt. Nevertheless, despite this potential mismatch, the ground truth strain of each of the walls visible in the synthetic images is made available in the open-access dataset.

Secondly, the CircAdapt strain patterns correspond to the myofibers located throughout the LV, but the orientations of the myofibers are not taken into account. Thus, we assumed the myofibers to be longitudinally distributed. Moreover, a pragmatic approach has been used to kinematically impose the CircAdapt strains on corresponding LV geometries. The apex was considered as the reference while estimating the longitudinal motion, and the epicardium was considered as the reference while estimating the radial motion. This resulted in a usual motion of the LV where the apex remains at rest in the longitudinal direction and the epicardium remains at rest in the radial direction.

The current implementation of our pipeline inherits typical echocardiographical artifacts, such as reverberation due to ribs and lungs, and clutter, from template clinical scans. Therefore, these artifacts cannot be independently added or removed in the simulated cardiac US data. Enhancing realism in terms of typical echographical artifacts could add value and impact the training performance of machine learning-based algorithms.

Regarding the impact of the aforementioned limitations on the simulated data, we found that the proposed assumptions may result in synthetic myocardial motions that do not exactly represent the corresponding CircAdapt strain patterns, particularly the regional strain patterns, which is beyond the scope of this work. Our approach, however, is sufficient for our purpose of generating a wide range of cardiac mechanics in large amounts that are realistic enough to aid the training of machine learning based motion estimation algorithms. With the availability of the RF data, scatter map and GT LV motions, the current large open-access dataset serves as a valuable resource for developing, improving and validating machine learning-based US data processing algorithms. These improved algorithms should, in turn, aid the cardiologist to make faster, and more accurate assessments.

V. CONCLUSION

We have presented a new realistic cardiac US simulation pipeline that enabled the generation of 2D synthetic cardiac US recordings on a large scale. Further, we also presented a large open-access database consisting of 1296 diverse, simulated, realistic cardiac US recordings, including 2-CH and 4-CH views, with diverse GT LV motion, geometry and US speckle patterns. This dataset can be accessed using the following link [36]: https://doi.org/10.48804/Q1VEX8.

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